with eigenvector w. Then $\alpha \pm \beta$ is an eigenvalue of

$$\mathcal{A} \otimes \mathrm{Id}_m \pm \mathrm{Id}_n \otimes \mathcal{B},$$

and $\alpha\beta$ is an eigenvalue of

$$\mathcal{A} \otimes \mathcal{B}.$$

The corresponding eigenvector is in both cases $v \otimes w$. This shows that both $\overline{\mathbb{Z}}$ and $\overline{\mathbb{Q}}$ are rings. To show that the latter is a field, it suffices to note that if $\alpha \neq 0$ is a root of a polynomial $p(x)$ of degree d, then α^{-1} is a root of $x^d p(1/x)$. The last statement is easy, since a number α is algebraic if and only if it defines a finite extension of $\mathbb{Q}$. $\square$

Proposition 5.2.5. $\overline{\mathbb{Z}} \cap \mathbb{Q} = \mathbb{Z}$.

Proof. We will be using Definition 5.2.1. Let z be a root of

$$p(x) = x^n + a_1 x^{n-1} + \ldots + a_{n-1}x + a_n,$$

and suppose

$$z = \frac{p}{q} \in \mathbb{Q}, \quad \gcd(p, q) = 1.$$

Notice that the leading term of $p(z)$ will have q^n in the denominator, whereas all the other terms will have a lower power of q there. Thus, if $q \neq \pm 1$, then $p(z) \notin \mathbb{Z}$, a contradiction. Thus, $z \in \overline{\mathbb{Z}} \cap \mathbb{Q} \Rightarrow z \in \mathbb{Z}$. The reverse inclusion follows because $n \in \mathbb{Z}$ is a root of $x - n$. $\square$

Every algebraic number α has a **minimal polynomial** $p(x)$ which is the monic polynomial with rational coefficients of the smallest degree such that $p(\alpha) = 0$. Any other polynomial $q(x)$ with rational coefficients such that $q(\alpha) = 0$ is divisible by $p(x)$. Roots of $p(x)$ are called the **algebraic conjugates** of α; they are roots of any polynomial q with rational coefficients such that $q(\alpha) = 0$.

Note that any algebraic conjugate of an algebraic integer is obviously also an algebraic integer. Therefore, by the Vieta theorem, the minimal polynomial of an algebraic integer has integer coefficients.

Below we will need the following lemma:

Lemma 5.2.6. *If $\alpha_1, \ldots, \alpha_m$ are algebraic numbers, then all algebraic conjugates to $\alpha_1 + \cdots + \alpha_m$ are of the form $\alpha'_1 + \cdots + \alpha'_m$, where α'_i are some algebraic conjugates of α_i.*